Milestone 2

1 Project Overview

The goal of this project is to build an interactive visualization that presents the spatio-temporal evolution of global natural disasters from 1975 to 2025. Rather than focusing on isolated events, our approach aims to present a dynamic narrative of global risk and trends.

We want to focus on three key dimensions: (1) the evolution of disaster types and their frequency, distinguishing between climate-related and geological events; (2) the spatial distribution of disasters, identifying how vulnerable regions have shifted or intensified over time; and (3) the relationship between disaster frequency and human impact, including deaths and economic losses.

To realize this, we want to design an interactive website that embodies the experience of visiting a museum. The visualization will follow a storytelling approach, in which users navigate through "exhibits" that progressively reveal different aspects of the data.

2 Implementation Plan

In this section, we describe the different independent pieces and sketches that make up our project.

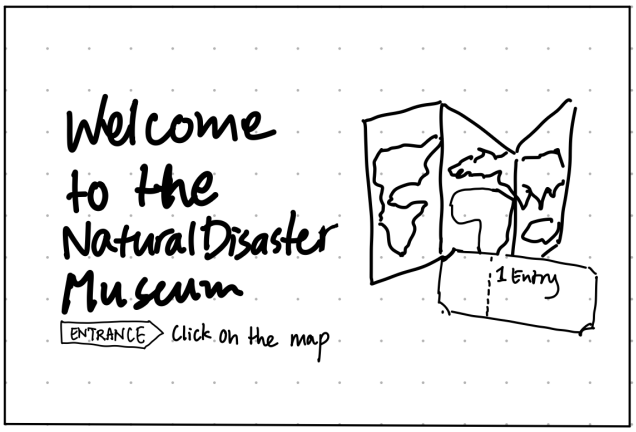
The visualizations are organized into three main layers: a **spatial layer**, which provides a global overview of disasters; a **temporal layer**, which allows users to explore country-specific statistics trends over time; and a **narrative layer**, which presents earth stories and insights.

Spatial Layer

2.1 Welcome Page

This is the first page that the users see when entering our website. To move on to the next, they are required to click on the map.

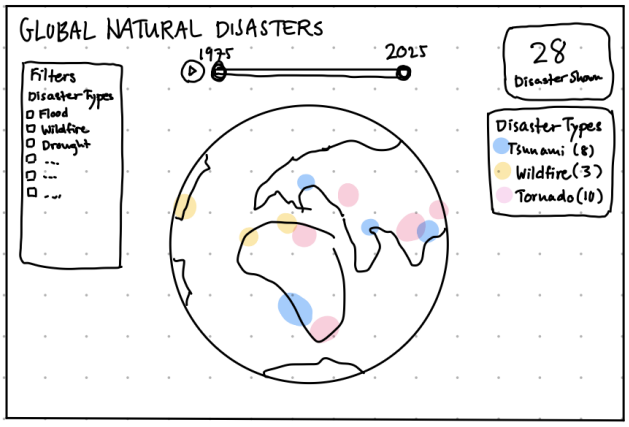
Sketch:



2.2 Main Page

This is our main page, displaying a 3D globe where each country is colored according to the number of recorded natural disasters. Users can interact with the globe (rotate, zoom) and apply filters based on disaster type (e.g., floods, earthquakes, storms). At the top center of the page, a temporal control bar allows users to explore the evolution of disasters over time across the globe. The user can manually adjust the time range using a slider.

Sketch:

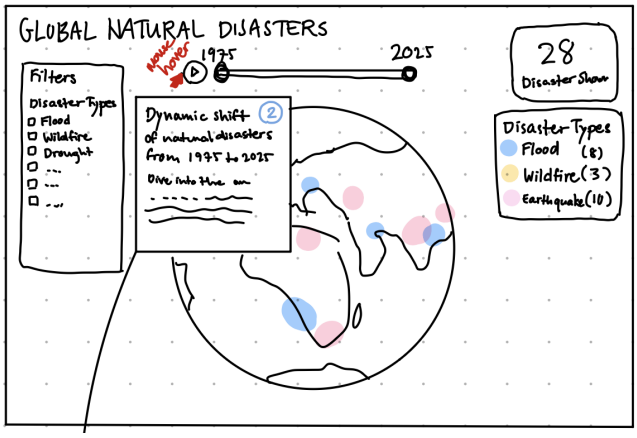


2.3 Play Button

The sketch illustrates what happens when a user interacts with the play button on the main page. When the user hovers over the button, a dialog appears explaining the feature.

Clicking the play button transitions the user into the **narrative layer** (see Section 2.7), where the visualization shifts from exploration to storytelling. Instead of only displaying a dynamic overview of disasters over time, this interaction guides the user through a sequence of curated insights and events.

Sketch:

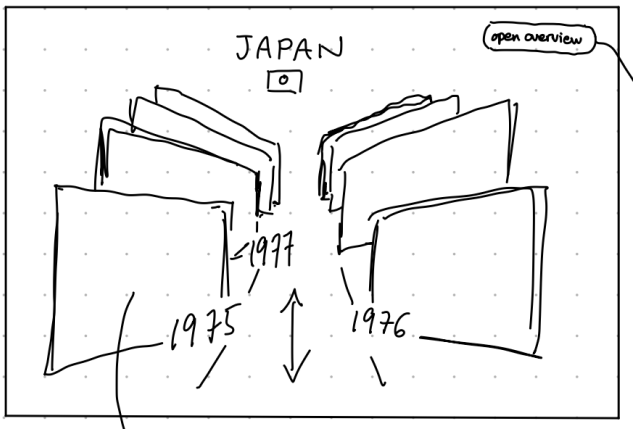


Temporal Layer

2.4 Country Timeline

This page appears when a user clicks on a specific country on the globe from the main page. It is designed as a museum-like experience, where users scroll vertically through a sequence of cards arranged from left to right, each representing a specific year. As users scroll, they move forward in time. Clicking on a card reveals detailed statistics on natural disasters for the selected country in that year. A button in the top-right corner directs the user to the country overview page, described in Section 2.6.

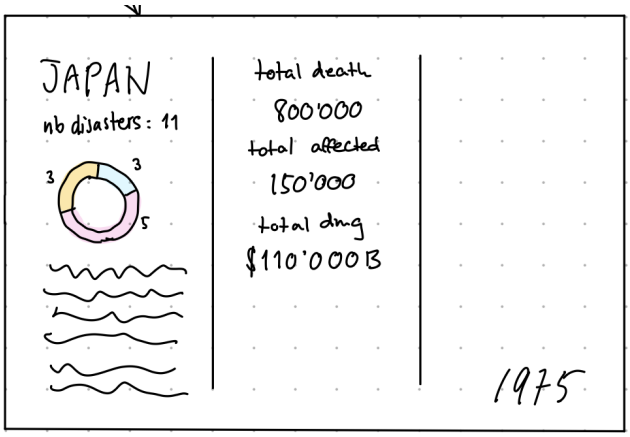
Sketch:



2.5 Country Statistics by Year

This page appears when a user clicks on a specific card in the country timeline. It displays a pie chart showing the distribution of natural disaster types for that year, along with key statistics such as the total number of deaths, the number of people affected, and the overall damage costs.

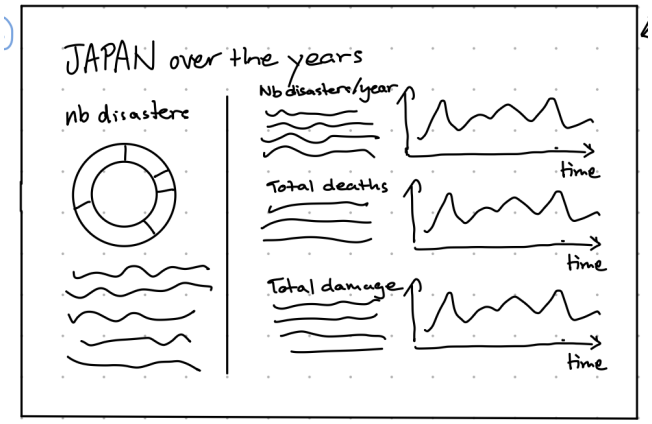
Sketch:



2.6 Country Overview

This page appears when the user clicks the "open overview" button on the country timeline page. It displays statistics on natural disasters over time for a specific country, including a pie chart showing the distribution of disaster types and line charts illustrating the evolution of total disasters, deaths, and overall damage over the years.

Sketch:



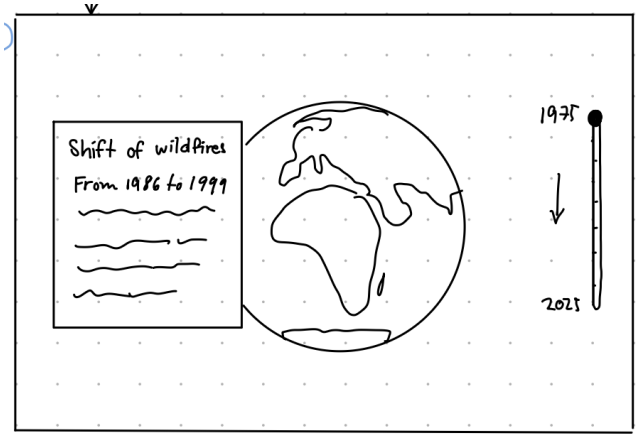
Narrative Layer

2.7 Earth Stories

This section represents the core of the narrative layer. Its goal is to provide a storytelling-driven exploration of natural disasters, allowing users to move through time and discover key events, impactful disasters, and significant trends.

This part is inspired by the experience of visiting a museum, the interface presents a sequence of curated "stories" that appear as the user progresses through time. Navigation is designed to be both guided and interactive. Users can scroll through the interface to move forward in time and progressively reveal new stories, or directly interact with a timeline positioned on the right side. This timeline can be dragged manually, allowing users to explore specific periods.

Sketch:



3 Tools and Lectures

Visualization	**Tool(s)**	**Relevant Lecture(s)**
Welcome Page	D3.js / TopoJSON / world-atlas@2 / SVG / CSS	1.1 / 1.2 / 4.2 / 8.2
Main Page	Three.js / globe.gl / D3.js / TopoJSON / world-atlas@2 / WebGL	1.1 / 1.2 / 4.2 / 8.2 / 12.2
Country Timeline & Overview	D3.js / SVG / CSS	1.1 / 1.2 / 4.2 / 5.2
Earth Stories	D3.js / SVG	1.1 / 1.2 / 4.2 / 5.2

4 Extra Ideas

As an extra and more ambitious idea, we would like to add a **country comparison race plot**, animating the evolution of the type, number of natural disasters and economic losses across different countries over the years. This would offer a dynamic temporal perspective and let viewers observe how patterns of disaster frequency and economic impact shift globally over time. This enhancement enriches the project beyond its core message, and could be dropped if time constraints arise.

A second optional extension would be a **Switzerland-focused zoom-in**, incorporating a dedicated Swiss dataset to provide a more detailed view of natural disasters at the national level. This would let users explore local trends more precisely, adding regional relevance to the project. Since the main visualization focuses on the globe, this remains a supplementary feature that can be omitted without affecting the overall project.